

MONISH DHILIPKUMAR

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EDUCATION

George Mason University

Fairfax, Virginia

Masters of Science in Computer Science; GPA: 3.63

January 2023 - December 2024

Coursework: Software Architecture, UI/UX, Agile Methodologies, Operating Systems, Embedded Systems, Analysis of Algorithms

Anna University

Chennai, India

Bachelors in Computer Science and Engineering

April 2018 - March 2022

Coursework: Data Structures, Cloud Computing, Artificial Intelligence, Object Oriented programming, Cryptography and Network Security

PROFESSIONAL EXPERIENCE

DLK Technologies Ltd

May 2022 - August 2022

Software Engineer Intern

Remote

- Developed full-stack components for 3+ web applications, reducing loading times by 25% and improving adherence to design principles, which led to a 30% increase in user engagement and satisfaction.
- Collaborated effectively with cross-functional teams on test development and debugging for an online car auction platform, aligning project requirements with business goals through clear communication and interpersonal skills. Applied expertise in Java, C++, JavaScript, HTML, CSS, and SQL to solve complex technical challenges.
- Integrated and orchestrated 5+ RESTful APIs to streamline third-party data workflows, improving system interoperability by 50%. Applied strategic debugging techniques, reducing issue resolution times by 35%.
- Optimized real-time data handling and scalability by designing and deploying containerized microservices using Docker and Kubernetes, ensuring high availability during peak traffic.
- Conducted weekly updates and code reviews, analysing over 10+ submissions per week to improve quality, maintainability, and adherence to coding standards, boosting overall team efficiency by 30%.
- Enhanced data processing pipelines by implementing SQL query optimization and indexing strategies, resulting in a 25% improvement in database performance and faster data retrieval for high-concurrency systems.
- Built reusable UI components with React.js, improving development speed and ensuring consistency across multiple web applications.
- Automated testing workflows using industry-standard frameworks, increasing test coverage by 40% and ensuring robust application functionality across various environments.
- Implemented caching mechanisms with Redis to reduce API response times, improving system performance and end-user experience.
- Developed and deployed monitoring tools to track key performance indicators, reducing time to detect and resolve system anomalies by 30%.
- Participated in Agile workflows, including daily stand-ups, sprint planning, and retrospectives, ensuring alignment with team objectives and timely delivery of features.
- Worked with version control systems like Git to maintain and review codebase changes, ensuring seamless collaboration and version consistency across the team.
- Resolved 20+ critical production issues using advanced debugging techniques in both front-end and back-end systems.

ACADEMIC PROJECTS

Online Grocery Shopping Website

- Revamped and launched an online grocery shopping platform, coding inventory management and streamlining order processes, which increased operational efficiency by 40%.
- Created an AI-powered chatbot system using TensorFlow, enhancing user experiences and a 20% boost in customer engagement.
- Integrated secure payment gateways and real-time analytics dashboards to monitor system performance, reducing downtime by 30%.

Diabetes Detection Using Deep Learning Algorithms

- Built a diabetes detection model using CNNs and KNN algorithms, leveraging Jupyter for development and improving model accuracy by 15%.
- Applied advanced pre-processing techniques to normalize a dataset of 10,000+ records, increasing robustness and reducing time by 25%.
- Deployed the model via a Flask-based web interface, enabling real-time predictions for end-users and expanding usability.

Computer Science Department Website

- Engineered a responsive website using React, HTML5, CSS, and Bootstrap, focusing on intuitive navigation and user-friendly layouts.
- Reduced backend query response times by 40% by integrating C++ functionalities into a Spring Boot backend.
- Established an automated CI/CD pipeline using Git and Jenkins, ensuring efficient deployments and enhanced team collaboration.

Employee Task Management System

- Designed and implemented front-end components using React.js, with backend logic developed in Node.js and TypeScript.
- Deployed containerized microservices on AWS with Kubernetes, improving scalability and increasing uptime by 25%.
- Integrated task performance analytics using Scikit-learn, providing actionable insights to optimize team workflows.

Intelligent Expense Management System

- Developed an expense tracking application with Python, integrating machine learning models for predictive spending analysis.
- Optimized RESTful APIs and SQL queries for backend operations, ensuring smooth functionality under high-concurrency scenarios.
- Created a React-based user interface with a focus on accessibility and intuitive navigation, enhancing the overall user experience.

SKILLS AND CERTIFICATIONS

Programming: Java, C/C++, JavaScript, Python, PHP

Frameworks and Tools: Django, MySQL, Git, GitLab, Kafka, Spring boot

Expertise: Web Development (UI/UX, HTML, CSS), Debugging (WinDBG, GDB), Testing, Database (MySQL, schemas)

Cloud & DevOps: AWS (S3, EC2, lambda), Azure, GCP, CI/CD pipelines, Docker, Kubernetes, Jenkins.

Microservices: Deployment and containerization, RESTful API integration, and scalable service architecture

Database Management: MySQL, PostgreSQL, NoSQL, MongoDB

Certifications: SAA-C03 - AWS Certified Solutions Architect Associate

Building Web Applications in PHP, HTML and CSS – Coursera